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1. Introduction

Timely forest monitoring, with easily comprehensible data analysis and outputs, is growing increasingly urgent. Since 2015, the Sentinel-1 mission's satellites have allowed for unprecedented, regular observation of tropical forests at a 6-, 12-, or 24- day interval. C-band synthetic aperture radar (SAR) scenes are acquired day and night, regardless of cloud cover, haze, smoke, or aerosols. Sentinel-1 Ground Range Detected (GRD) data, with its capability to provide such information in all-weather conditions, is a powerful resource for monitoring land cover changes, including potential deforestation. The use of Sentinel-1 backscatter images for deforestation detection comes from analyzing changes in the Earth's vegetated surface that occur due to the removal of forest cover (i.e. removal of vegetation components of trees). This data is particularly valuable in tropical regions where cloud cover can hinder the use of optical-based satellite datasets. The challenge, however, is in finding adequate methods to extract meaningful 'indicators' of forest loss from the vast amount of incoming SAR data, such that anomalies in the time-series can be regularly and consistently detected across tropical forests. Further, such forest-monitoring methods should be transparent and easily understandable to the wider public, hence enabling confidence in their use across various public and private sectors.

Under the main Sentinel-1 for Science Amazonas contract funded by ESA since 2020, using Sentinel-1 Synthetic Aperture Radar (SAR) backscatter (σ^0) time series, a complete workflow was developed for detecting forest change using a space-time data cube architecture ("StatCubes") and a multi-temporal change detection (MCD) approach. The system included automated cloud-based preprocessing of Sentinel-1 GRD scenes on the Copernicus CreoDIAS platform, comprising calibration, terrain correction using the SRTM DEM, and speckle filtering. Twelve-day SAR mosaics were generated across the full Amazon basin, and statistical indicators—such as backscatter differences, variability, trend metrics, and simple significance tests—were extracted to form the StatCubes used for detecting potential forest loss. A transparent thresholding and decision-tree method was then applied to identify change from 2017–2021, supported by forest/non-forest and water masks derived from 2015–2016 Sentinel-1 data and external datasets including Global Forest Watch and SRTM-based hydrological masks. The main contract delivered basin-wide forest-change maps, a public viewing portal, and open-source tools to ensure reproducibility.

The Change Request (CCN) introduces important methodological and operational extensions to this baseline system. In addition to the statistical and threshold-based workflow, new AI- Convolutional Neural Networks (CNNs) based models specifically for semantic segmentation of satellite imagery were incorporated to better capture complex patterns in the Sentinel-1 time series. These models complement the StatCube indicators and improve detection robustness, particularly for subtle or heterogeneous disturbance events. The CCN also re-implements the full processing chain—preprocessing, StatCube extraction, statistical MCD, and AI inference—within the openEO federation [1] on the Copernicus Dataspace Ecosystem [2]. This enables transparent, scalable, and backend-agnostic execution across cloud environments. All resulting algorithms have been published in the ESA APEX Algorithm Catalogue to support discoverability, reuse, and alignment with ESA's broader application ecosystem. Finally, both the classical MCD outputs and the new AI-enhanced change layers have been integrated into the EarthCODE environment.

2. Overview of the Statistics-based MCD methodology

The statistics-based Multi-temporal Change Detection (MCD) methodology identifies forest loss by analysing deviations in the temporal behaviour of Sentinel-1 SAR backscatter at the pixel level. The approach relies on a set of statistical descriptors computed from multi-temporal SAR mosaics and applies transparent, rule-based criteria to detect change events and assign them to specific dates. The method was first prototyped and upscaled on the [CreoDIAS cloudferro cloud-services](#) [3]. Detailed description of this approach can be found in the ATBD v1 [9].

2.1 Input Data and Temporal Configuration

The method operates on Sentinel-1 GRD backscatter mosaics generated at a 12-day interval for VV and VH polarisations. Data from 2015–2016 are used to establish baseline forest and non-forest conditions, after which forest-loss detection is performed from 2017 onwards. Time-series analysis is carried out using moving temporal windows, allowing the method to assess changes relative to local historical context rather than long-term averages.

2.2 Feature Extraction (StatCubes)

For each temporal window, a set of statistical descriptors—referred to as *StatCubes*—is computed to characterise short-term behaviour in the SAR signal. These descriptors include backscatter differences between past and future observations, measures of variability, linear trend parameters, and statistical significance tests comparing distributions across the window. Together, the StatCubes provide a compact representation of temporal change behaviour for each pixel and time step. Each test produces an independent indication of disturbance, and candidate change events are defined where several indicators concur within the same temporal window.

2.3 Decision-Tree Integration and Masking

The outputs of the individual statistical tests are combined using a rule-based decision-tree framework. A pixel is classified as changed only when a minimum number of StatCube indicators exceed predefined thresholds. Base-year forest/non-forest masks and water masks derived from Sentinel-1 backscatter are applied, together with external reference layers such as Global Forest Watch [4] and SRTM-based hydrological masks, to restrict detections to valid forest areas. Iterative masking using past and future windows further reduces transient artefacts.

2.4 Outputs

The statistics-based MCD methodology produces (i) 12-day forest-loss detection layers, (ii) a cumulative multi-year forest-loss map.

3. Overview of the AI-based MCD methodology

The analysis of backscatter time series using AI involves designing the U-net architecture with additional layers specific to this segmentation task, creation of training data, training the model, and finally prediction. These steps are explained in the following sections.

3.1 Semantic segmentation model

Convolutional Neural Networks (CNNs) are composed of multiple layers that each perform specific functions to process and learn from data. The foundational layers of a CNN include:

3.1.1. Convolutional Layers

These layers apply filters to the input images to create feature maps, highlighting important features such as edges, textures, or shapes.

3.1.2. Activation Layers

Following convolution, activation functions like ReLU (Rectified Linear Unit) introduce non-linearity, helping the network learn complex patterns.

3.1.3. Pooling Layers

Pooling (usually max pooling) reduces the dimensionality of the feature maps, making the model more efficient and less prone to overfitting by summarizing the features.

3.1.4. Fully Connected Layers

These layers, present towards the end of the CNN, perform classification based on the features extracted and pooled from the previous layers.

3.1.5. Softmax or Sigmoid Layer

This is typically the final layer, used for making predictions. Softmax is used for multi-class classification, while sigmoid is used for binary classification.

3.1.6. U-Net Architecture for Semantic Segmentation

The U-Net architecture is a CNN variant designed specifically for semantic segmentation tasks, where the goal is to classify each pixel of an image into a specific category (illustrated in Figure 1). It has a distinctive "U-shaped" structure that consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. The innovation of U-Net lies in its use of skip connections, which directly concatenate feature maps from the contracting path to the expanding path. This approach helps the network to recover the spatial information lost during downsampling.

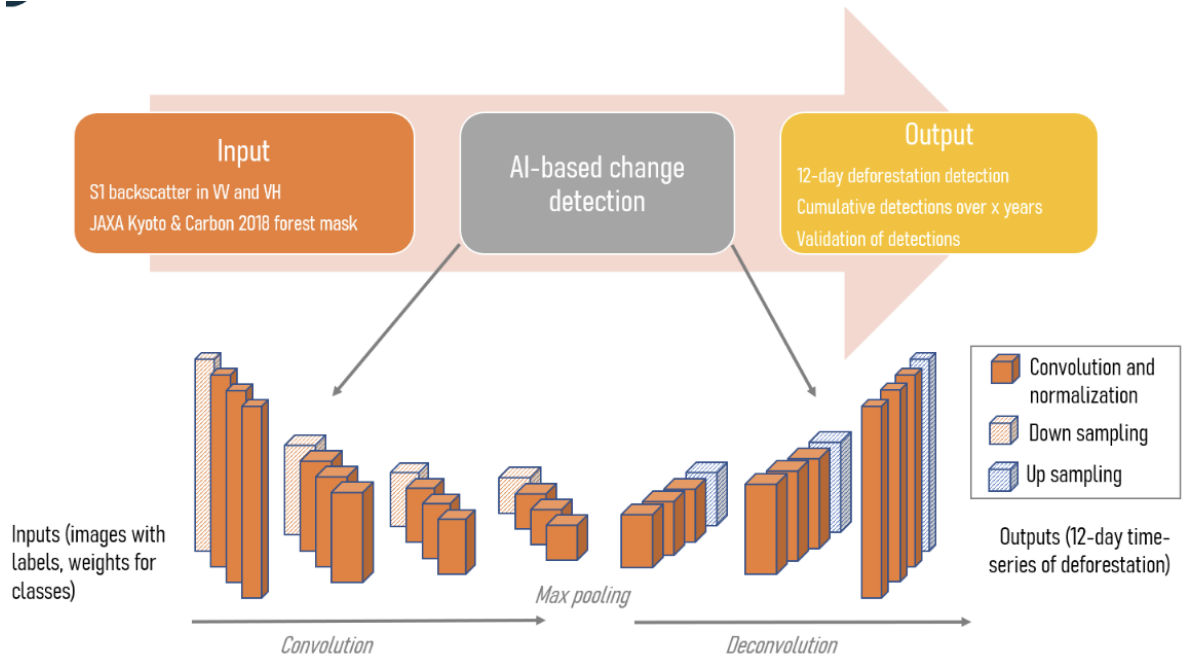


Figure 1. U-net architecture for the AI-based MCD (figure reproduced from [5]).

3.2 Preparation of training dataset

3.2.1. Labels

The training dataset was built from two complementary label sources: the threshold-based Multi-temporal Change Detection (MCD) workflow and Global Forest Watch (GFW) tree-cover loss data. From the MCD system, the 12-day change layers served as the main source of positive examples, capturing short-term deforestation signals identified through multiple statistical tests and filtered by forest and water masks. To expand geographic coverage and avoid overfitting to MCD-specific patterns, additional labels were derived from GFW, which provided independent optical-based loss detections. Areas classified as intact forest in both GFW and the MCD base-year masks were used as negative samples. Together, the MCD and GFW labels yielded a more balanced and representative dataset, enhancing the model's ability to generalise across different disturbance patterns.

3.2.2. Backscatter images

The Sentinel-1 GRD scenes that processed to sigma-naught backscatter images (currently available as a collection on openEO, see Figure 2) are used as training data. For a given area and given time period, polarized backscatter images of VV, VH, and VH/VV bands are stacked. From this a 3-band geoTIFF is created. Multi-temporal mosaics are created every 12 days using Sentinel-1 SAR backscatter (σ^0) data. These mosaics include data for two types of polarizations (VV and VH) and two orbital directions (ascending and descending, when possible), and the ratio band (VH/VV) is computed from these mosaics. Within the openEO platform processed earth observation data for the specific area and time period can be extracted. In some cases, backscatter images will have no-data pixels. When a particular pixel has some no-data value, values are corrected by going back in time.

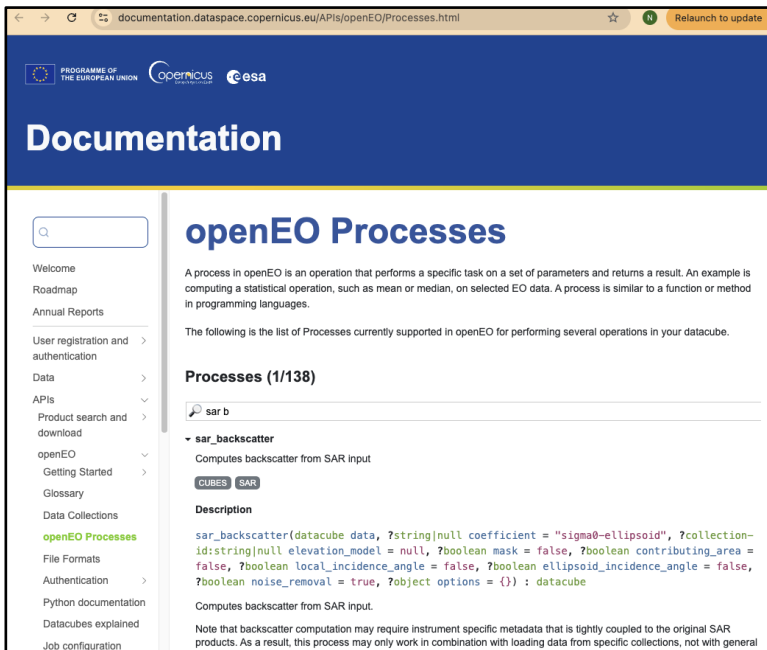


Figure 2: A process in openEO that performs a specific task on a set of parameters and returns a result.

3.3 Prototyping

Within the AI-based architecture, a moving time window is used to detect deforestation. The CNN operates on spatial patches extracted from each Sentinel-1 mosaic, with patch sizes of either 256×256 or 512×512 pixels. These dimensions were chosen to balance two requirements: (1) patches must be large enough to capture the spatial context of typical deforestation patterns—such as clearings, edges, or clustered disturbances—which often span tens to hundreds of metres; and (2) they must remain small enough to fit efficiently into GPU memory, allowing the model to process long temporal stacks without excessive computational overhead. The 256×256 size provides sufficient spatial structure for training while enabling a large number of samples.

For each time window, 256×256 patches are extracted from the sequence of SAR mosaics, forming stacks of size $256 \times 256 \times n$, where n is the number of mosaics included in the temporal window. The length of the moving time window is selected to capture the duration over which deforestation-related backscatter changes remain detectable, as these disturbances typically persist across several acquisition cycles. Using multiple consecutive mosaics provides the temporal depth needed for the CNN to identify consistent change patterns rather than isolated fluctuations. With this configuration, the 17 Sentinel-2 tiles processed over two years yield approximately 80,000 stacked samples, each paired with its corresponding label.



Figure 3. Selected 17 (approximately 100 x 100 km) Sentinel-2 tile extents chosen for prototyping.

The AI model produces a continuous probability map with values between 0 and 1, where higher values indicate intact forest and values closer to 0 reflect a higher likelihood of forest loss. As deforestation occurs, the model output for affected pixels decreases toward 0. A threshold is then applied to convert this probability map into a binary forest-loss layer (see Figure 4). A pixel is classified as forest loss when its probability value falls below 0.2. This threshold was determined empirically by comparing model outputs with reference labels and testing cut-off values between 0.05 and 0.5. A value of 0.2 provided the best balance between limiting false positives and reliably capturing true deforestation across the study areas.

During upscaling, output probabilities from the AI model are scaled by a factor of 10,000 and stored as integers to reduce file size and streamline processing. In this scaled format, the operational threshold corresponds to a value of 2000 (equivalent to a probability of 0.2). The threshold, however, is not fixed and can be modified easily [in the code](#) to suit different applications, confidence requirements, or local calibration efforts.

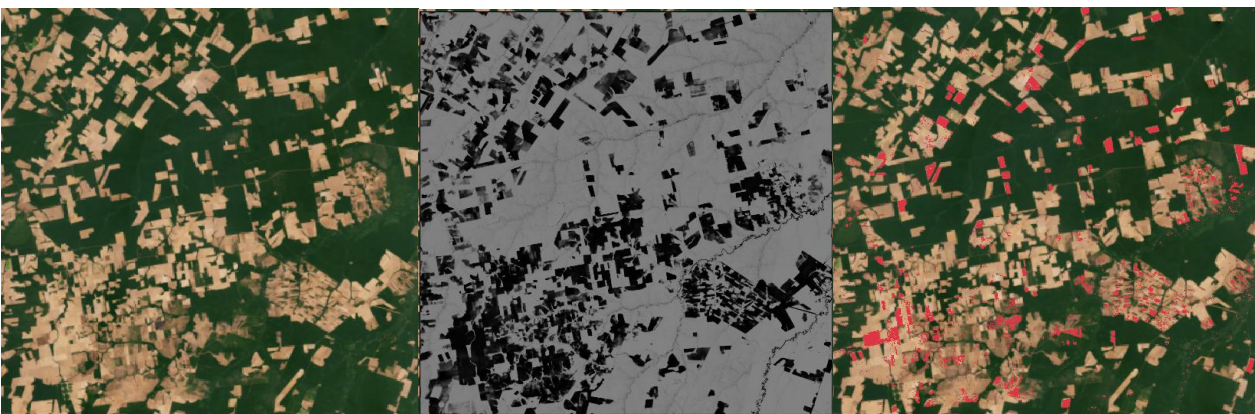


Figure 4: (Left) Planet Lab imagery from April 2024, (center) probability of change detection from AI-based MCD algorithm, and (right) changes classification after application of thresholds and apply relevant masks.

2. Spatial Filtering, harmonization and temporal consolidation

Initial forest-loss detections are generated using the statistics-based MCD workflow by applying thresholds to multiple StatCube descriptors derived from Sentinel-1 backscatter time series. A pixel is considered a candidate change event only when a minimum number of these indicators exceed predefined thresholds within the same moving temporal window. This step produces a preliminary set of 12-day/24-day binary forest-loss detection layers. To complement the statistical change detections, an AI-based change-prediction layer is generated using the convolutional neural network (CNN) to capture spatially coherent disturbance patterns that may not be fully identified by the threshold-based MCD approach alone.

AI-derived detections may be temporally offset from statistically detected change events, a temporal harmonisation step is applied. AI detections are matched to statistical MCD detections occurring within a ± 60 -day window. When detections from both methods overlap spatially within this temporal window, they are treated as a single forest-loss event and merged, with the event date assigned to the nearest statistical MCD detection. Spatial consistency is evaluated using 8-connected neighbourhood connectivity, whereby pixels sharing a common edge or corner are considered adjacent. AI detections that are spatially adjacent to statistically detected change pixels within the harmonisation window are retained, enabling the extension of change regions along forest edges or within fragmented clearings. In contrast, AI detections that lack both temporal proximity and spatial adjacency to statistical MCD detections are discarded, thereby reducing false positives as illustrated in Figure 4. The code snippet from [6] is shown here:

```
ai_detection_timewindow = find_cumulative_aidetection(ai_files_list)
ai_mcd_detection = create_ai_mcd_merged_detection(mcd_array,
ai_detection_timewindow)
```

Following the fusion of statistical and AI-based detections, a forest/non-forest mask derived from base-year Sentinel-1 data and external reference layers is applied to restrict the results to valid forest areas. A spatial filtering step is then performed using connected-component analysis (e.g. GDAL *gdal_sieve* [7]) to remove small, isolated pixel clusters below a minimum size threshold.

The refined detections are consolidated across the full monitoring period to generate a cumulative forest-loss product. A spatial filtering step is then performed to remove excessively large contiguous regions that are characteristic of non-deforestation processes such as flooding or scene-boundary artefacts. This cumulative product is used to enforce spatial and temporal consistency in the individual 12-day outputs, retaining only change detections that persist coherently over time and suppressing temporally inconsistent or spurious signals.

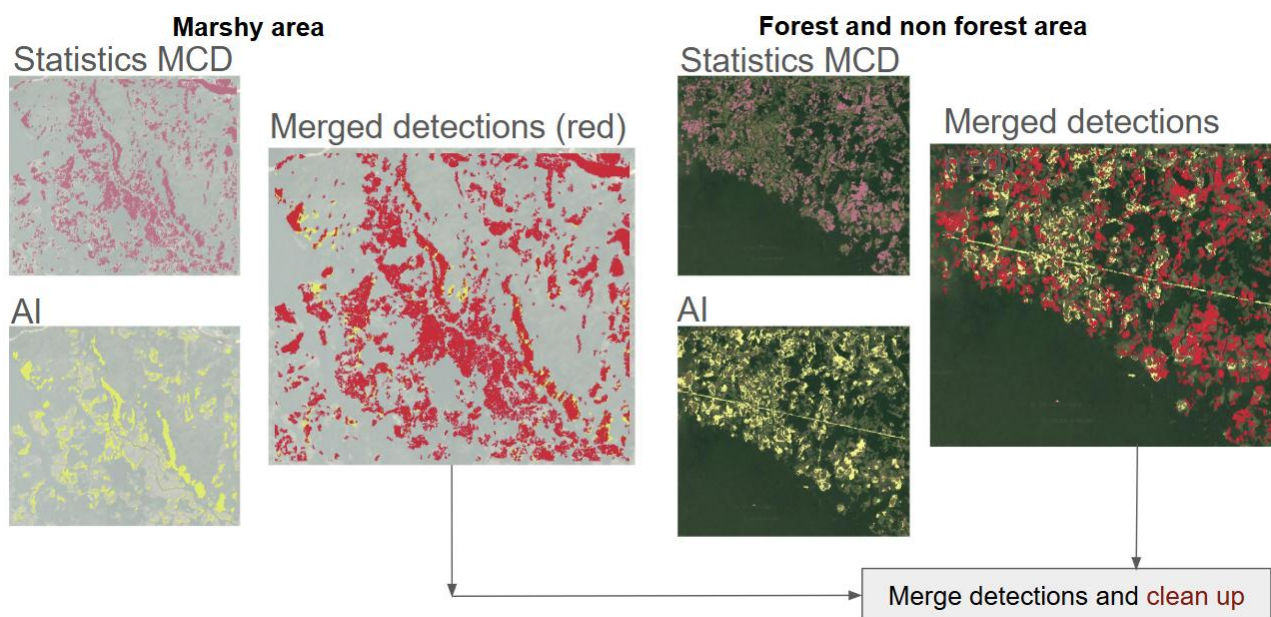


Figure 4: Two examples of deforestation detection over the extent of a Sentinel-2 tile (namely, 18NYH). The 'merged detections' contain both the threshold-based and AI-based results (seen in red). Other potential false alarms are discarded (e.g. those seen in yellow)

3. Transfer to openEO for operation

The decision was taken to move the processing of the statistics-based and AI-based MCD to openEO, with the support of ESA. Specific to the AI-based methodology for transfer to openEO - once a CNN segmentation model is trained, typically on a dedicated local machine or cloud environment, the model needs to be exported in *onnx* format, which is compatible with the openEO backend. After deploying the model to openEO, it can be used to process satellite imagery directly on the platform. Users can specify their area of interest and time range to retrieve backscatter data from Sentinel-1 SAR imagery or other relevant datasets available through openEO. The model then applies its learned segmentation patterns to this data, predicting deforestation. The predictions can be extracted as raster data which are useful for further analysis. By leveraging openEO's infrastructure, the process of applying CNN models to satellite data becomes streamlined and scalable, allowing for efficient analysis of large geographical areas over extended periods.

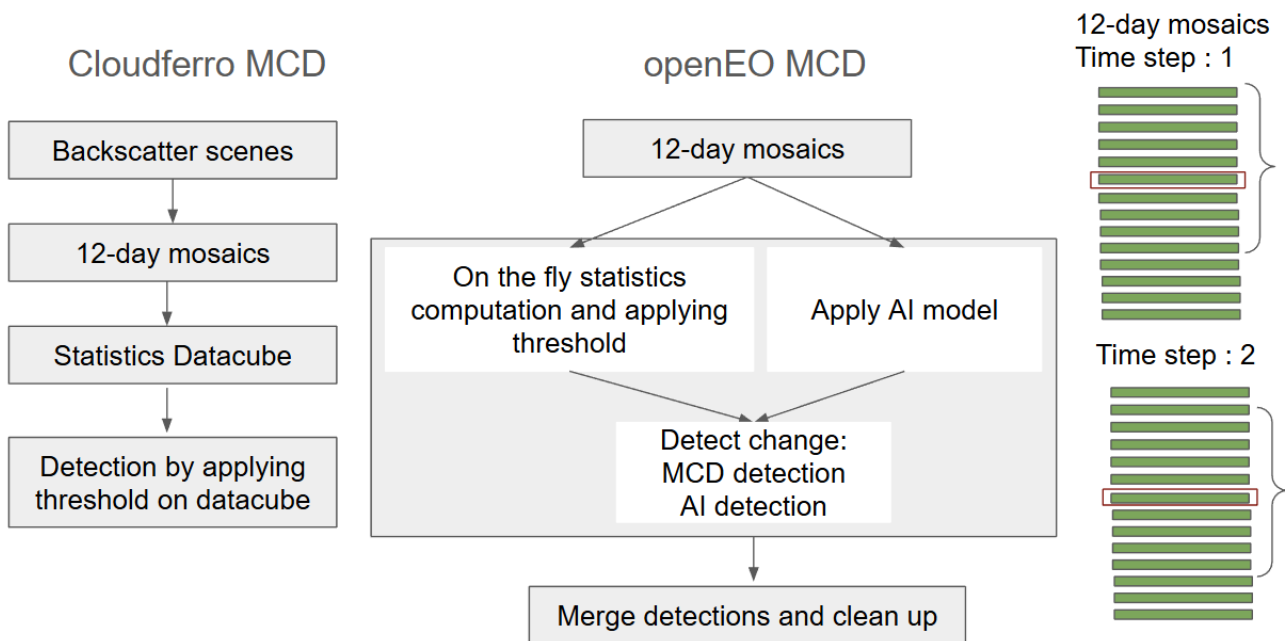


Figure 5: Schematic illustration of the processing of backscatter scenes, creation of mosaics, and implementation of Sentinel 1 for Amazonas deforestation detection algorithm.

The workflows of both the original threshold-based MCD and the AI-based MCD are now fully implemented on openEO (see Figure 5). Processing begins with the backscatter images, where the script can compute ten 12-day backscatter values on the fly using [openeo sarbackscatter process](#).

```
load_collection_options = {"polarisation": lambda pol: pol == "VV&VH"}

s1_collection = connection.load_collection('SENTINEL1_GRD', bands=['VV', 'VH'],
    properties=load_collection_options, spatial_extent=bbox)
    .filter_temporal(extent=master_temporal_extent)

s1_backscatter = s1_collection.sar_backscatter(
    coefficient="sigma0-ellipsoid",
    elevation_model="COPERNICUS_30",
    local_incidence_angle=False)

s1_backscatter = s1_backscatter.resample_spatial(resolution=20, method="near")
```

```
# Apply temporal aggregation over intervals
s1grd = s1_backscatter.aggregate_temporal(
    intervals=temporal_extents,
    reducer="mean")
```

□

It should be noted that at the time of upscaling and data processing, backscatter was generated using sigma0-ellipsoid. This process does not perform terrain correction and speckle filtering. During prototyping and testing, backscatter generated from ESA SNAP [8] was used.

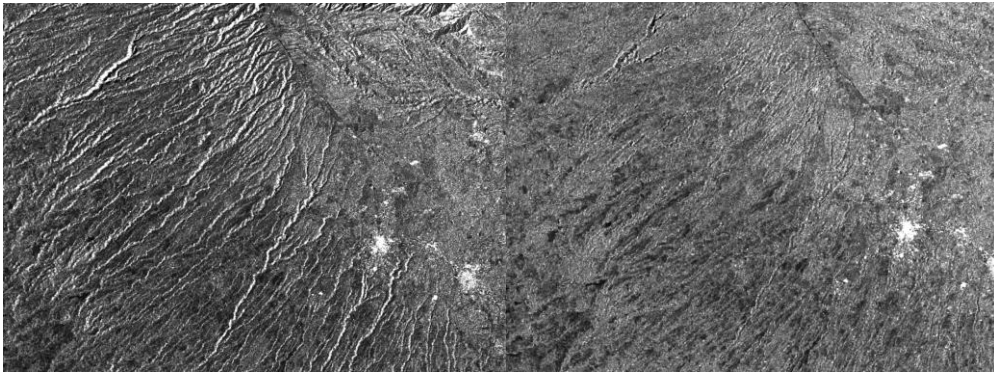


Figure 6: (left) Backscatter from openEO sarbscatter process using sigma0-ellipsoid without terrain correction and speckle filtering. (right) Backscatter from SNAP using sigma0 with terrain correction and speckle filtering.

If precomputed backscatter images are available, then they can be uploaded to s3 storage. A STAC can be created from these images and openEO [can retrieve precomputed backscatter data](#) from STAC.

```
□bucket_name = "mys3bucket"
catalog_relative_path =
    "mybackscatterfolder/mybackscatterfolder_catalog.json"
url = f"https://s3.waw3-1.cloudferro.com/swift/v1/{bucket_name}
   }/{catalog_relative_path}"

stac_item = connection.load_stac(
    url=url,
    spatial_extent=aoi,
    temporal_extent=temporal_extent,
    bands=bands
)
```

The Sentinel-1 input data are first harmonised in space and time to enable consistent multi-temporal analysis. All scenes are resampled to a common spatial resolution of 20 m and temporally aggregated over predefined monitoring intervals using a mean reducer. Linear interpolation is applied along the temporal dimension to address irregular acquisition gaps, after which the backscatter values are transformed to the logarithmic (decibel) domain to stabilise variance and enhance separability of forest and non-forest signals.

```

datacube = stac_item.resample_spatial(resolution=20,
method=resample_method)
datacube = datacube.aggregate_temporal(intervals=temporal_extents,
reducer="mean")
datacube = datacube.apply_dimension(dimension="t",
process="array_interpolate_linear")
datacube = datacube.apply(lambda x: 10 * eop.log(x, 10))

```

The temporally aggregated time series is then collapsed into a multi-band representation by mapping each time window to a dedicated band for both VV and VH polarisations, preserving temporal order through systematic band naming.

Reduce to a single array by collapsing the time dimension into a single stack

```

datacube_time_as_bands = datacube.apply_dimension(
    dimension='t',
    target_dimension='bands',
    process=lambda d: eop.array_create(data=d)
)
band_names = [band + "_t" + str(i+1).zfill(2) for band in ["VV", "VH"]
for i in range(len(temporal_extents))]
datacube_time_as_bands = datacube_time_as_bands.rename_labels('bands',
band_names)

```

Statistical change detection is subsequently performed using a user-defined function operating on the temporally stacked bands. In parallel, an AI-based change-detection module is applied using a sliding-window neighbourhood strategy, enabling the detection of spatially coherent disturbance patterns. The outputs from the statistical and AI-based analyses are finally merged into a unified data cube, providing complementary per-time-window change indicators that form the basis for subsequent [harmonisation, filtering, and generation of forest-loss products](#).

Load the UDF from a file.

```

udf = openeo.UDF.from_file(Path(__file__).parent.resolve() /
"udf_mcd_deforestation_detection.py")

```

Apply the UDF to the data cube.

```

datacube_udf = datacube_time_as_bands.apply_dimension(
    process=udf,
    dimension="bands"
)
target = [band + "_t" + str(i+1).zfill(2) for band in ["MCD",
"MCDthreshold", "VVpmin", "VHpmin"] for i in
range(number_of_timewindows)]
datacube_udf = datacube_udf.rename_labels(dimension="bands",

```

```

target=target)

udf_ai = openeo.UDF.from_file(Path(__file__).parent.resolve() /
"udf_ai_deforestation_detection.py")
# Apply the UDF to the data cube.
datacube_ai_udf = datacube_time_as_bands.apply_neighborhood(
    process=udf_ai,
    size=[
        {"dimension": "x", "value": 192, "unit": "px"},
        {"dimension": "y", "value": 192, "unit": "px"},
    ],
    overlap=[
        {"dimension": "x", "value": 32, "unit": "px"},
        {"dimension": "y", "value": 32, "unit": "px"},
    ])
target = [band + "_t" + str(i + 1).zfill(2) for band in ["AIMCD"] for i
in range(number_of_timewindows)]
datacube_ai_udf = datacube_ai_udf.rename_labels(dimension="bands",
target=target)
change_detection = datacube_udf.merge_cubes(datacube_ai_udf)

```


4. Access to methodology of scripts on Github

The scripts used for creating multi-temporal mosaics from processed backscatter data and for detecting tree cover change are publicly accessible via Gisat's GitHub repository at <https://github.com/gisat/S14amazonas>. This repository provides open access to the data, methodologies, and scripts involved in the process.

Within the repository, the **master branch** contains the original MCD workflow in its complete form. In addition, a dedicated branch named **openeo** has been created, where the entire processing chain has been migrated to the openEO technology on the CDSE. This branch also incorporates job management executed on the openEO infrastructure as well as the post-processing of results.

Furthermore, another branch, titled **ai_cnn**, hosts scripts developed and refined during the prototyping phase of the AI-based MCD. This branch contains numerous intermediate scripts that were iteratively written, tested, and re-tested. Together, these scripts offer valuable insights into the experimental and developmental stages of the workflow.

Overall, the repository serves not only as a resource for replicating the operational workflows but also as a record of the methodological evolution from prototyping to platform integration.

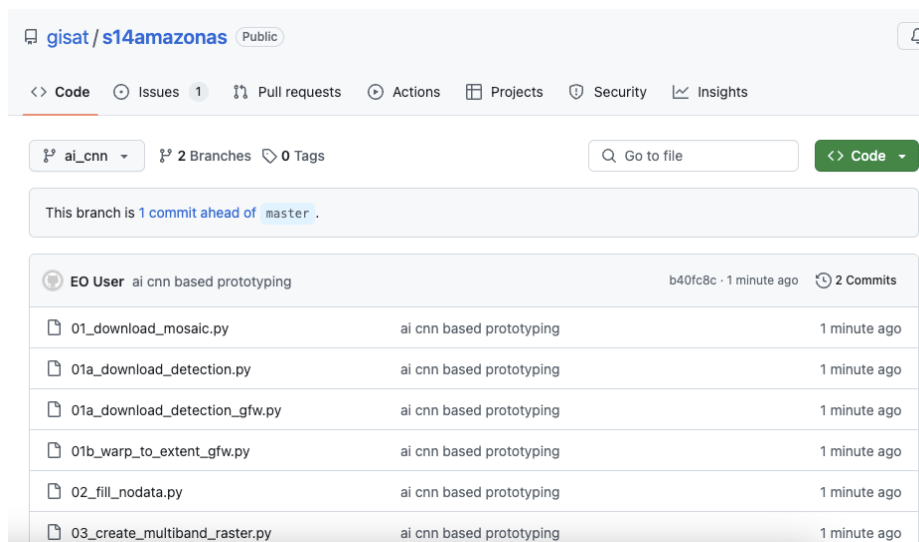


Figure 7. Snapshot of the Github repository with codes for the AI-based change detection

5. Computational requirements and up-scaling the MCD

Significant optimizations were introduced during the upscaling of the MCD based on statistics and AI workflow to improve both computational efficiency and cost-effectiveness. Since the workflow relies on applying a moving time window over a sequence of backscatter images, the initial approach of launching individual tasks for each window proved highly resource-intensive. To address this, the workflow was restructured so that multiple time windows could be processed in batches. By loading a given set of backscatter data once and executing several detection runs concurrently, redundant data transfers and repeated processing were minimized. This batching strategy substantially reduced overall resource consumption while maintaining analytical accuracy and scalability for large-scale deforestation monitoring.

To demonstrate the data production capabilities, a representative set of tiles was selected, covering diverse biomes across the Amazon Basin and Central America. In total, approximately 165 tiles were processed, ensuring ecological variability and geographical coverage. The change detection workflow was executed over a four-year period, providing a robust temporal dataset for assessing forest dynamics and validating the scalability of the methodology.

Processing a single Sentinel-2 tile extent over a four-year period, required the computation of backscatter and the execution of the detection script, amounting to approximately 1,000 credits in the openEO platform. This credit-and-slot model provides a flexible and scalable mechanism to balance throughput, cost efficiency, and operational predictability for large-scale geospatial processing. This difference in the backscatter generation, resulted in the false positives and noisy detections, especially in regions with terrain change such as hilly regions.

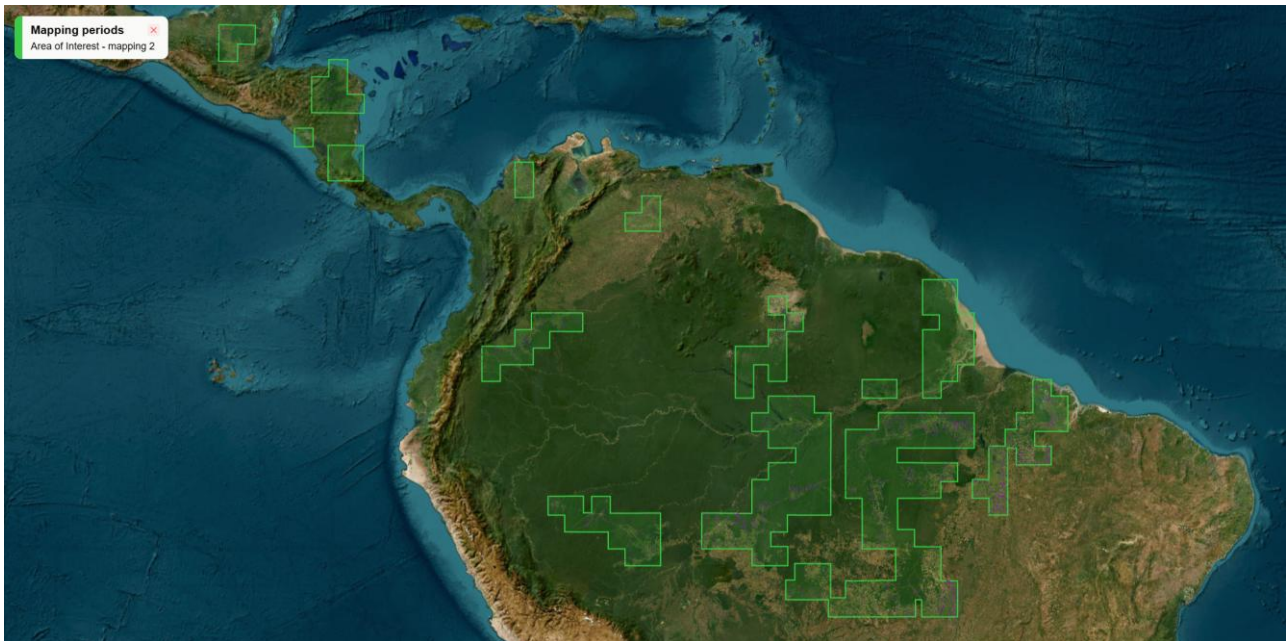


Figure 8: Selected 165 tiles for the demonstration of the MCD on openEO.

To assess the applicability of the developed workflow beyond the Amazon Basin, the method was also tested in selected regions of Central America. These experiments were conducted using the same statistical and AI-based configurations as in the Amazon case. The resulting change maps were generated following the identical processing chain.

It should be noted that the test regions exhibit pronounced terrain variability. In the initial experiments, non-terrain-corrected Sentinel-1 backscatter without speckle filtering generated using openEO technology in CDSE backend was used, which led to strong terrain-induced backscatter effects and a high level of noise in the change detections. As illustrated in Figure 9, the use of non-terrain-corrected backscatter significantly alters the signal behaviour, resulting in excessive false detections in areas of complex topography.

To investigate this effect, an additional test was performed using terrain-corrected backscatter with speckle filtering generated in ESA SNAP. The subsequent application of the change-detection workflow showed a substantial reduction in noise and produced more spatially consistent forest-change and forest-loss detections. A second round of validation, focusing on areas previously affected by terrain-induced artefacts, confirmed a clear improvement in detection quality. These results corroborate that the methodology performs significantly better when terrain-corrected backscatter is used, particularly in regions with complex topography.

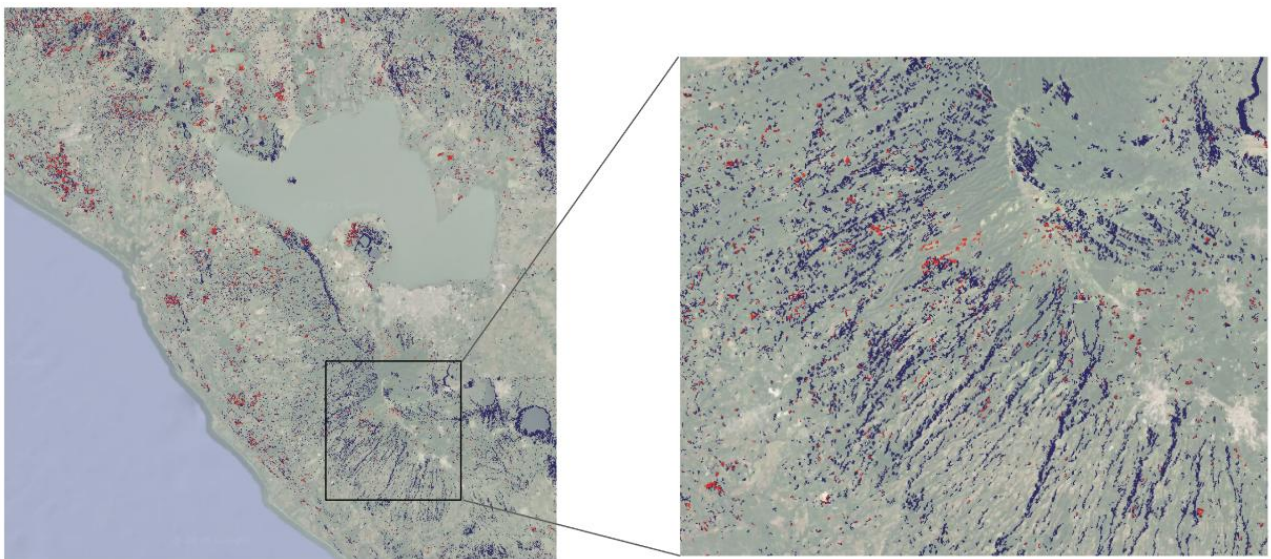


Figure 9: Example of tree-cover-change detections over a Sentinel-2 tile extent (16PEU) in a region with complex terrain. The left panel shows the full tile extent, while the right panel provides a zoomed view of the highlighted area. Detections derived from the change-detection workflow are overlaid on the background imagery, illustrating the impact of terrain-induced backscatter effects when non-terrain-corrected Sentinel-1 data are used (blue), which results in elongated and noisy detection patterns aligned with topography. The detections from terrain corrected and speckle filtered backscatter images are in red.

6. References

- [1] <https://openeo.org/>
- [2] <https://dataspace.copernicus.eu/ecosystem/services/openeo>
- [3] <https://cloudferro.com/cases/creodias/>
- [4] <https://www.globalforestwatch.org/dashboards/global/>
- [5] Ronneberger, O., Fischer, P., Brox, T. (2015). U-Net: Convolutional Networks for Biomedical Image Segmentation. In: Navab, N., Hornegger, J., Wells, W., Frangi, A. (eds) Medical Image Computing and Computer-Assisted Intervention – MICCAI 2015. MICCAI 2015. Lecture Notes in Computer Science(), vol 9351. Springer, Cham. https://doi.org/10.1007/978-3-319-24574-4_28
- [6] https://github.com/Gisat/S14amazonas/blob/openeo/openeo_postprocess2_treecoverchange.py
- [7] https://gdal.org/en/stable/programs/gdal_sieve.html
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- [9] D7.1_S14AMA_MCD_AlgorithmTheoreticalBasisDocument_v1.pdf